MIL-P-12741B(MU)
17 MARCH 1964

SUPERSEDING
MIL-S-12741A(Ord)
23 OCTOBER 1961

MILITARY SPECIFICATION

# PROJECTILE, 4.2 INCH ILLUMINATING, M335A1 METAL PARTS, FOR

## 1. SCOPE

**1.1** This specification covers Projectile, 4.2 Inch Illuminating, M335A1 Metal Parts.

## 2. APPLICABLE DOCUMENTS

**2.1** The following documents of the issue in effect on date of invitation for bids or request for proposal, form a part of this specification to the extent specified herein.

### SPECIFICATIONS

MILITARY

MIL-A-2550 —Ammunition and Special Weapons, General Specification for.
MIL-P-10025 —Packing and Marking for Domestic Shipment of Inert Ammunition Components, General Specification for.
MIL-I-45208 —Inspection Systems Requirements.

### STANDARDS

FEDERAL

Federal Test Method Standard No. 151 —Metals; Test Methods

MILITARY

MIL-STD-105 —Sampling Procedures and Tables for Inspection by Attributes.
MIL-STD-109 —Quality Assurance Terms and Definitions.
MIL-STD-639 —Surface Roughness Values Preferred for Army Ordnance Materiel.
MIL-STD-1235 —Single and Multilevel Continuous Sampling Procedures and Tables for Inspection by Attributes.

### DRAWINGS

MUNITIONS COMMAND

F10521880 —Projectile, 4.2 Inch Illuminating, M335A1 Metal Parts.
F8848553 —Pallet.
IEL-10521880 —Inspection Equipment List for Projectile 4.2 Inch Illuminating, M335A1 Metal Parts.

(Copies of specifications, standards, drawings, and publications required by suppliers in connection with specific procurement function should be obtained from the procuring activity or as directed by the contracting officer.)

## 3. REQUIREMENTS

**3.1 General.** Materials, parts, and assemblies shall comply with the requirements specified on the applicable drawings, referenced specifications and the following:

**3.2 Initial production sample.** Unless otherwise directed by the contracting officer, an initial production sample shall be required.

**3.3 Heat-treatment or stress relief.** Heat-



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treatment or stress relief applied shall be uniform throughout each lot of components and shall be so conducted as to produce in any component the minimum mechanical properties specified on the applicable drawing. No forming of metal involving plastic flow without subsequent stress relief is permitted.

**3.4 Mechanical properties.** Mechanical properties required by the applicable drawings shall be uniform throughout each component and meet the minimum specified properties.

**3.5 Metal defects.** All components shall be free from cracks, seams, splits, cold shuts, inclusions, porosity or similar defects.

**3.6 Brazed joint.** Braze metal shall be visible for the full 360 degrees around the exterior diameters of the body—adapter joint.

**3.6.1** *Sectionalized joint.* When sectionalized, the braze metal joint shall be free from voids, cracks, entrapped slag for a minimum of 85% of the exposed joint area. Presence of cleaning or other fluids in the solder groove shall be considered evidence of a defective joint.

**3.7 Functional.**

**3.7.1** *Security—Initial production sample.* There shall be no breakup, deformation or evidence of failure of the body assembly in the weapon's bore or in flight.

**3.8 Workmanship.** The requirements for workmanship are as shown on the applicable drawings, in the applicable specifications and the requirements of MIL–A–2550, and the following:

**3.8.1** *Surface finish.* The requirements for surface finish are as detailed on the applicable drawings. The cavity shall be free of burrs, scale fins, deep seams or die scratches, pits or other surface discontinuities.

**3.8.2** *Protective coating.* The protective coating shall be in accordance with the assembly drawing and the specifications listed thereon.

**3.8.3** *Threads.* Threads shall be full and undamaged for the entire minimum length or depth as required by the applicable drawing and shall be in accordance with the provisions of MIL–A–2550.

**3.8.4** *Burr.* No part shall have a burr which might interfere with the assembly or function of the round or might be injurious to personnel handling the item.

**3.8.5** *Foreign matter.* No part or assembly shall contain dirt, grease, chips, rust, corrosion or other foreign matter.

## 4. QUALITY ASSURANCE PROVISIONS

**4.1 Responsibility for inspection.** Unless otherwise specified in the contract or purchase order, the supplier is responsible for the performance of all inspection requirements specified herein. Except as otherwise specified, the supplier may utilize his own facilities or any commercial laboratory acceptable to the Government. The Government reserves the right to perform any of the inspections set forth in the specification where such inspections are deemed necessary to assure supplies and services conform to prescribed requirements.

**4.1.1** *Quality assurance terms and definitions.* Reference shall be made to MIL–STD–109 to define quality assurance terms used.

**4.1.2** *Inspection.* Inspection shall be in accordance with MIL–A–2550 and MIL–I–45208 except as specified herein.

**4.2 Initial production sample.** Prior to initiation of regular production the contractor shall submit an initial production sample to a designated Government approved facility for evaluation. The sample shall consist of 40 metal parts assemblies as follows: 30 completely assembled including paint; 5 assembled unpainted; 5 sets component parts unassembled and unpainted. The sample shall be subjected to the tests listed in 4.3.4. The sample shall be manufactured in the same manner, using the same materials, equipment, procedures as will be used in regular production. All parts and materials including packaging and packing must be the same as will be used for regular production and shall be obtained from the same source of supply. The contractor shall inspect the sample in accordance with his inspection system and the requirements of



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Specification MIL–I–45208. After inspection and provisional acceptance at source, the initial production sample shall be inspected at the Government approved facility for all requirements of the drawings and specifications. The Government reserves the right to require new initial production samples until such time as an acceptable sample is submitted. Twenty-five (25) of the metal parts assemblies, 15 stamped R and 10 stamped S shall be forwarded to a designated proving ground for ballistic testing in accordance with 4.3.5 utilizing the procedure in 4.4.4.

4.2.1 *Disposition of sample.*

(a) 25 of the completely assembled and painted metal parts, 10 stamped S as directed by the procuring activity and 15 stamped R, shall be forwarded by the contractor to a designated proving ground for ballistic testing in the phases indicated in 4.3.5.1.

(b) The 15 remaining samples shall be forwarded to a designated Government approved facility for evaluation.

**4.3 Inspection provisions.**

4.3.1 *Lot.*

4.3.1.1 *Submission of product.* The product shall be submitted in accordance with MIL–STD–105 or MIL–STD–1235, as applicable.

4.3.1.2 *Tubing lot.* A lot of tubing from which bodies are to be manufactured shall not contain more than one heat of steel.

4.3.1.3 *Forging lot.* A lot of base plug forgings shall not contain more than one heat of steel.

4.3.1.4 *Cold shaping lot.* Cold shaping lots shall contain not more than one heat of steel, except in those instances where the manufacturer, with prior approval of the procuring agency has demonstrated that his process is capable of producing acceptable components from steel within a specified chemistry range.

4.3.2 *Examination.* One hundred percent examination shall be performed for all critical defects. Examination for major and minor defects shall be performed on a defect basis in accordance with the classification of defects contained herein, using the sampling plans contained in MIL–STD–105 or MIL–STD–1235 as applicable. An AQL of 0.25 percent for each major defect and an AQL of 0.40 percent for each minor defect shall be used. All non-conforming material shall be rejected. Four power (4X) magnification may be used in performing visual examination for such defects as burr and foreign matter. The Ordnance Standard Finish Blocks prescribed by MIL–STD–639 shall be used as a basis of comparison for surface roughness determination.

4.3.3 *Classification of defects.*

4.3.3.1 *Spacer, Drawing C10521881 covering a detail of Drawing F10521880.*

| | *Defect* | *Method of Inspection* |
|---|---|---|
| CRITICAL. None defined. | | |
| MAJOR | | |
| 101. | Diameter outside, small | Gage |
| 102. | Length of small diameter, min. | Gage |
| 103. | Surface finish on small diameter | Visual See 6.2 |
| 104. | Metal defective | Visual |
| MINOR | | |
| 201. | Diameter of holes | Gage |
| 202. | Diameter outside, large | Gage |
| 203. | Length total | Gage |
| 204. | Location of holes | Gage |
| 205. | Thickness of wall | Gage |
| 206. | Radius of chamfer, missing or incorrect | Visual See 6.2 |
| 207. | Surface finish, improper | Visual See 6.2 |
| 208. | Burr | Visual |



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4.3.3.2 *Adapter, nose Drawing D10521883 covering a detail of Drawing F10521880.*

| | *Defect* | *Method of Inspection* |
|---|---|---|
| CRITICAL. None defined. | | |
| MAJOR | | |
| 101. | Concentricity of spacer cavity with brazing area including perpendicularity of front shoulder | Gage |
| 102. | Concentricity of small inside diameter with spacer cavity | Gage |
| 103. | Diameter of spacer cavity | Gage |
| 104. | Diameter of brazing area | Gage |
| 105. | Surface finish of spacer cavity improper | Visual See 6.2 |
| 106. | Metal defective | Visual |
| MINOR | | |
| 201. | Diameter of large shoulder | Gage |
| 202. | Diameter of inside small | Gage |
| 203. | Depth of solder groove, max. | Gage |
| 204. | Depth of spacer cavity | Gage |
| 205. | Length from front shoulder to front of shoulder groove, min. | Gage |
| 206. | Length from front shoulder to rear of solder groove, max. | Gage |
| 207. | Length from front face to front shoulder | Gage |
| 208. | Length of brazing area | Gage |
| 209. | Length total | Gage |
| 210. | Radii or chamfer missing or incorrect | Visual See. 6.2 |
| 211. | Burr | Visual |
| 212. | Surface finish improper | Visual and Tactual |

4.3.3.3 *Adapter Assembly, Drawing D10521882 covering a detail of Drawing F10521880.*

| | | |
|---|---|---|
| CRITICAL. None defined. | | |
| MAJOR | | |
| 101. | Length total, max. | Gage |
| 102. | Gap between spacer and spacer seat | Visual See 6.2 |
| MINOR | | |
| 201. | Assembly damaged | Visual |
| 202. | Foreign matter | Visual |

4.3.3.4 *Plug, Base (prior to painting) Drawing D10521885 covering a detail of Drawing F10521880.*

| | | |
|---|---|---|
| CRITICAL. None defined. | | |
| MAJOR | | |
| 101. | Concentricity of large outside diameter with face of flange including perpendicularity of front face flange | Gage |
| 102. | Concentricity of pitch diameter of thread with press fit area, including perpendicularity of front and rear faces of flange. | Gage |
| 103. | Summetry of rotating disc seat with major diameter of thread. | Gage |
| 104. | Diameter of force fit area, at rear of taper | Gage |
| 105. | Diameter of flange | Gage |
| 106. | Width of rotating disc seat | Gage |
| MINOR | | |
| 201. | Diameter thread undercut | Gage |
| 202. | Depth of cavity | Gage |
| 203. | Height of rotating disc seat | Gage |
| 204. | Height of thread boss | Gage |
| 205. | Length of tapered area | Gage |
| 206. | Length total | Gage |
| 207. | Width of thread undercut, max. | Gage |
| 208. | Thickness of flange | Gage |
| 209. | Thickness of wall at front of cavity | Gage |
| 210. | Radii of chamfer missing or incorrect | Visual See 6.2 |



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| | Defect | Method of Inspection |
|---|---|---|
| 211. | Surface finish improper | Visual See 6.2 |
| 212. | Burr | Visual |
| 213. | Foreign matter | Visual |

**4.3.3.5** *Plug, Base (after paint) Drawing D10521885 covering a detail of Drawing F10521880.*

CRITICAL. None defined.

MAJOR

| | | |
|---|---|---|
| 101. | Pitch diameter of thread, min. | Gage |
| 102. | Major diameter of thread, min. | Gage |
| 103. | Surface finish on force fit area, damaged | Visual |

MINOR

| | | |
|---|---|---|
| 201. | Pitch diameter of thread, max. | Gage |
| 202. | Protective coating inadequate or damaged | Visual |
| 203. | Wax emulsion inadequate or damaged | Visual |
| 204. | Corrosion preventive compound missing from thread | Visual |
| 205. | Foreign matter | Visual |

**4.3.3.6** *Body, Drawing D10521884 covering a detail of Drawing F10521880.*

CRITICAL. None defined.

MAJOR

| | | |
|---|---|---|
| 101. | Diameter of brazing area | Gage |
| 102. | Variation of wall thickness | Gage |
| 103. | Thickness of wall at ogive | Gage |
| 104. | Metal defective | Visual |

MINOR

| | | |
|---|---|---|
| 201. | Diameter at front, outside | Gage |
| 202. | Length to datum diameter | Gage |
| 203. | Length to rear of front bourrelet | Gage |
| 204. | Length to start of rear bourrelet | Gage |
| 205. | Length total | Gage |
| 206. | Surface finish improper | Visual and Tactual |
| 207. | Burr | Visual |
| 208. | Foreign matter | Visual |

**4.3.3.7** *Metal Parts Assembly (prior to paint) Drawing F10521880.*

CRITICAL. None defined.

MAJOR

| | | |
|---|---|---|
| 101. | Concentricity of pitch diameter of nose thread with bourrelets including perpendicularity of front face of nose. | Gage |
| 102. | Concentricity of nose adapter with bourrelets | Gage |
| 103. | Concentricity of forward position of ogive with bourrelets. | Gage |
| 104. | Concentricity of rear position of ogive with bourrelets. | Gage |
| 105. | Concentricity of body with bourrelets | Gage |
| 106. | Concentricity of illuminant charge cavity with bourrelets | Gage |
| 107. | Concentricity of base plug with the bourrelets | Gage |
| 108. | Diameter of bourrelet(s), min. | Gage |
| 109. | Diameter of body | Gage |
| 110. | Diameter of illuminate charge cavity | Gage |
| 111. | Depth of thread cavity | Gage |
| 112. | Perpendicularity of rear face with bourrelets | Gage |
| 113. | Braze metal not visible for full 360 degrees around exterior of body and adapter joint | Visual |
| 114. | Mating surfaces at body, adapter joint improper | Visual See 6.2 |
| 115. | Failure to pass load test by absence of letter "P" on body | Visual |



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| | Defect | Method of Inspection |
|---|---|---|
| MINOR | | |
| 201. | Diameter of nose outside | Gage |
| 202. | Diameter of thread counterbore | Gage |
| 203. | Depth of thread counterbore | Gage |
| 204. | Ogive over-laps nose adapter | Visual See 6.2 |
| 205. | Radii or chamfers incorrect | Visual See 6.2 |
| 206. | Burr | Visual |
| 207. | Foreign matter | Visual |

4.3.3.8 *Metal Parts Assembly (after paint) Drawing F10521880.*

CRITICAL. None defined.

| | | |
|---|---|---|
| MAJOR | | |
| 101. | Weight | Approved Scale |
| 102. | Pitch diameter of nose thread, max. | Gage |
| 103. | Minor diameter of nose thread, max. | Gage |
| 104. | Diameter of bourrelets, max. | Gage |
| 105. | Diameter of base plug cavity | Gage |
| 106. | Depth of base plug cavity | Gage |
| 107. | Length of effective nose thread, min. | Gage |
| 108. | Surface finish on walls of base plug cavity damaged | Visual See 6.2 |
| MINOR | | |
| 201. | Pitch diameter of nose thread, min. | Gage |
| 202. | Staking notches missing or incorrect | Visual See 6.2 |
| 203. | Paint in cavity not dry | Tactile |
| 204. | Protective coating inadequate or damaged | Visual |
| 205. | Thread damaged or not full | Visual |
| 206. | Marking missing, incorrect or illegible | Visual |
| 207. | Foreign matter | Visual |

4.3.3.9 *Pallet, Drawing F8848553.*

CRITICAL. None defined.
MAJOR. None defined.

| | | |
|---|---|---|
| MINOR | | |
| 201. | Pallet damaged | Visual |
| 202. | Sleeve missing | Visual |
| 203. | Improperly packed | Visual |
| 204. | Strapping missing, loose or broken | Visual and Manual |

4.3.4 *Inspection testing.* The following tests shall be performed in accordance with the provisions of 4.4 except as provided in MIL–I–45208.

4.3.4.1 *Mechanical properties.*

4.3.4.1.1 *Tension tests.* Two bodies from each tubing lot, after final heat-treatment, and two base plugs from each forging lot, after final heat-treatment shall be subjected to this test to determine compliance with the yield strength and elongation requirements of the applicable drawing. Failure of any test specimen to comply with the requirements shall be cause for rejection of the lot.

4.3.4.1.2 *Strength of brazed joint.* Each body assembly shall be subject to a mechanical load test to determine compliance with requirements of the applicable drawing. Failure of any assembly to comply shall be cause for rejection of the assembly. Body assemblies which comply with drawing requirements shall be stamped with the letter "P" as indicated on the assembly drawing.

4.3.4.1.3 *Sectionalized joint.* Two body assemblies prior to painting shall be selected from each lot of body assemblies and subjected to this test to determine compliance with 3.6.1. Failure of any segment of the joint to comply shall be cause for rejection of the lot.



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4.3.4.2 *Protective coating.* The number of specimens for protective coating tests shall be in accordance with the applicable Military Standard listed on the drawing. Failure to comply with the requirements shall be cause for taking the corrective action recommended therein.

4.3.5 *Functional.* The following proving ground tests will be performed by the Government.

4.3.5.1 *Security and recovery.* The quantities listed in Table I will be ballistically tested in the phases indicated. (See 4.4.4.)

TABLE I.

| *Phase* | *Initial Production Sample Quantity* |
|---|---|
| Security | 10 |
| Recovery | 15 |

Security—Failure of the initial production sample to comply with 3.7.1 will be cause for rejection of the sample.

Recovery—The functional tests fired for recovery are not acceptance tests, but the results will be scrutinized carefully by the Munitions Command to determine quality of ammunition produced. (See 6.1.)

4.3.6 *Inspection equipment.* Examination and test shall be made using equipment on IEL–10521880, except as provided in Specification MIL–I–45208.

**4.4 Test methods and procedures.**

4.4.1 *Mechanical properties.*

4.4.1.1 *Body.* From each body selected in accordance with 4.3.4.1.1, two tension test specimens conforming to a standard T type specimen as prescribed by Federal Test Method Standard No. 151 shall be taken longitudinally from each side of the body 180 degrees apart. The criteria for determining yield strength and elongation by the offset method shall be as prescribed by Federal Test Method Standard No. 151.

4.4.1.2 *Base plug.* From each base plug selected in accordance with 4.3.4.1.1, one tension test specimen conforming to a standard "R" type specimen as prescribed by Federal Test Method Standard No. 151 shall be taken from the center of the plug in a transverse direction normal to the centerline of the part. The criteria for determining yield strength and elongation by the offset method shall be as prescribed by Federal Test Method Standard No. 151.

4.4.1.3 *Strength of brazed joint.* In performing the load test, the body will be held by an approved holding fixture and the required load applied to the rear of nose adapter. No deformation or separation of joint should occur.

4.4.2 *Sectionalized joint.* The samples provided in accordance with 4.3.4.1.3 shall be sectionalized into 4 segments by removing the front portion of the assembly at least one inch to the rear of the brazed area. The portion that has been removed shall be cut into 4 segments and the joint visually examined to determine compliance with 3.6.1.

4.4.3 *Protective coating.* The procedures for testing the cleaning of components and assemblies and the application of protective coatings shall be in accordance with the applicable drawings and the applicable Military Standard listed thereon. The salt spray test shall be performed utilizing the equipment prescribed by Federal Test Method Standard No. 151.

4.4.4 *Functional.*

4.4.4.1 *Security.* The metal parts assemblies from the initial production sample stamped with the letter "s" inert loaded to service weight will be fired from a weapon for which the projectile is standard, the propellant charge will be adjusted to obtain 112 percent of the maximum rated pressure of the round in a new tube. Observation will be made to determine compliance with 3.7.1.

4.4.4.2 *Recovery.* The metal parts assemblies from the initial production sample, inert loaded to the service weight and utilizing the service charge, will be fired in a weapon for which the projectile is standard. At least 60 percent of the projectiles fired will be recovered and examined to determine:



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(a) If deformation occurred to the projectile while in the tube as evidenced by rifling marks extending more than 180 degrees around the circumference or by increase or decrease in the diameter of the projectile at bourrelet.
(b) If dishing of the base or upsetting occurred.
(c) If the shell were cracked or broken by other than impact.

## 5. PREPARATION FOR DELIVERY

### 5.1 Packaging.

5.1.1 *Level C.* Base Plugs Part No. 10521885 shall be packed in accordance with the requirements of Specification MIL–P–10025.

### 5.2 Packing.

5.2.1 *Level C.* Metal parts assemblies shall be palletized in accordance with Drawing F8848553.

5.3 Labeling and marking shall be in accordance with Specification MIL–P–10025.

5.4 Data card information shall be as specified in Specification MIL–A–2550.

## 6. NOTES

**6.1 Ballistic test.** Rounds fired for recovery are not acceptance tests but results will be carefully analyzed by the Munitions Command to determine quality of ammunition produced. Upon examination after recovery, evidence of body deformation, dishing or upsetting of the base, or breakup of the projectile by other than impact will result in immediate investigation by the Munitions Command.

**6.2 Visual examination qualification.** When compliance with the applicable requirements is in doubt as a result of visual examination, this characteristic may be measured or gaged to determine acceptability.

**Custodian:**

Army—MU

**Preparing activity:**

Army—MU

**Project No. 1315–A324**



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